EE5904/ME5404 Part II

Project 1: SVM for Classification of Spam Email Messages

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INTRODUCTION

The report shows the comparison of performance of Support Vector Machines (SVMs) for different kernels and different parameters. The kernels are differentiated into three different categories, which are:

- Hard margin with linear kernel
- Hard margin with polynomial kernel
- Soft margin with polynomial kernel

This report should be able to provide a good comparison and understanding about SVMs and the method to successfully design one. The software being used is MATLAB due to the availability of the Optimization Toolbox for quadratic programming and calculation of support vectors using it.

METHODOLOGY

- **Data pre-processing**
  Different sets of data can represent the same pattern or feature. The method to recognize and extract such patterns or features involves pre-processing. This includes various other methods including scaling and standardization/normalization. The choice of pre-processing method does not have an effect on the overall outcome since no information is lost in the process. This report uses the standardization technique, deducting the mean of each column and dividing by the standard deviation. The mean and standard deviation of the training dataset is reused for the testing and evaluation datasets.
  $$Standardized\ data = (Data - Mean)/(Standard\ Deviation)$$
- **Verification of admissibility of kernels**
  Kernels are classification parameters used to find optimal hyperplanes for high dimensional data. They are used to transform linearly inseparable data into separable data. It becomes important to choose a suitable kernel to be able to do so. As mentioned before, three categories of kernels are first experimented with in the report.
  - Hard margin with linear kernel $K(x_1, x_2) = x_1^T x_2$
  - Hard margin with polynomial kernel $K(x_1, x_2) = (x_1^T x_2 + 1)^p$
  - Soft margin with polynomial kernel $K(x_1, x_2) = (x_1^T x_2 + 1)^p$
    [p takes the values 2, 3, 4, 5]

  The difference between hard and soft margins is the upper bound during selection of support vectors. For hard margins, the upper bound does not exist, and for soft margins, the upper bound is close to 0 for a larger margin with a potential risk of misclassification of data.

  The admissibility of kernels is essential in the process of calculation of support vectors. This is checked through Mercer's condition, where the eigenvalues of the kernel matrix must all be

greater than or equal to zero. If the condition is satisfied, an optimal hyperplane exists to classify new data.

- **Calculation and selection of support vectors**
  MATLAB has the Optimization toolbox which is very useful in performing quadratic programming to calculate the support vectors by using the *quadprog* function. The optimization settings have to be also set for calculation. The number of iterations to find the optimal hyperplanes are set in order to increase the efficiency of the code and save time. The support vectors are calculated until the minimum of objective function is found that satisfies the constraints and tolerances or the number of iterations reach the set limit. In the case of the latter, the optimal hyperplane is not found.
  After the calculation of support vectors, selection is based on margin limits set by the user for the respective hard or soft margins. The suitable support vectors are used to generate the discriminant function using the weights and bias and calculate the accuracy of the classification.

RESULTS & DISCUSSIONS

Table 1 Admissibility of kernels and existence of optimal hyperplanes for different kernels and parameters

Kernel	Admissibility of kernels				Optimal hyperplane			
Hard margin with linear kernel	Yes				Yes			
Hard margin with polynomial kernel	p = 2	p = 3	p = 4	p = 5	p = 2	p = 3	p = 4	p = 5
	Yes	Yes	No	No	Yes	Yes	Yes	No
Soft margin with polynomial kernel	C=0.1	C=0.6	C=1.1	C=2.1	C=0.1	C=0.6	C=1.1	C=2.1
p = 1	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
p = 2	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
p = 3	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
p = 4	No	No	No	No	Yes	Yes	Yes	Yes
p = 5	No	No	No	No	Yes	Yes	Yes	Yes

Table 2 Training and testing accuracies for different kernels and parameters

Kernel	Training Accuracy (%)				Testing Accuracy (%)			
Hard margin with linear kernel	93.95				92.77			
Hard margin with polynomial kernel	p = 2	p = 3	p = 4	p = 5	p = 2	p = 3	p = 4	p = 5
	99.85	99.90	99.90	99.95	91.02	90.43	89.78	90.23
Soft margin with polynomial kernel	C=0.1	C=0.6	C=1.1	C=2.1	C=0.1	C=0.6	C=1.1	C=2.1
p = 1	93.05	84.60	83.10	82.55	92.32	84.24	82.88	81.97
p = 2	98.40	84.90	83.05	81.20	92.06	82.23	81.70	79.43
p = 3	70.95	71.50	70.50	68.15	69.79	69.99	68.75	67.45
p = 4	77.90	64.55	63.50	63.25	75.32	63.35	62.30	61.78
p = 5	48.65	49.80	50.03	51.40	48.89	50.06	50.85	51.76

Key takeaways:

- **Hard margins with linear kernels produced a good level of training and testing accuracy. The accuracy was above 90% without a high degree of overfitting. The kernel was admissible, and the algorithm converged to a minimum.**

- Hard margins with polynomial kernels showed a varied performance. For small values of p (2 and 3), the kernels were admissible and optimal hyperplanes existed. The training accuracy, however, reached up to over 99%, which signifies overfitting. The testing accuracy dropped, compared to the linear kernel. For p = 4, the kernel was not admissible but the algorithm converged. This may be due to a well separable training dataset that does not affect the classification to a large extent. However, the performance may be limited to new data. For p = 5, the kernel was not admissible and the number of iterations reached the maximum limit set and the performance is similar to when p = 4. The same conclusion can be derived here, and the performance can not be ensured for unseen data.
  A higher degree of kernel is not suitable for filtering of spam data. The training accuracies reach almost 100% and the testing accuracies reach about 90%. This shows that there is overfitting and generalization is possible to a certain extent only.
- Soft margins with polynomial kernels also failed to provide very promising results. When p = 1, it is a soft margin with a linear kernel, and the performance can be regarded as good since the training accuracy does not imply overfitting and the testing accuracy is very close. However, the accuracies drop for increasing values of C, i.e., the cost of violating constraints. For a second degree polynomial, the training accuracy is quite high, which could possibly mean overfitting, and the testing accuracy is also lower. For higher degrees, the performance is worse and the accuracies are even lower.
  Linear kernels with higher soft margins provide promising results with high training and testing accuracies and lower possibility of overfitting.

DESIGN DISCUSSIONS

According to the above listed takeaways, an SVM to filter spam emails should perform better with linear kernels. In the case of completely separable data, hard margins are said to perform better. In the current case of high dimensional data, it is difficult to predict whether the dataset is completely separable. A larger margin will be preferred to allow any slight changes in the features and better detection of spam messages. **Hence, a soft margin with a linear kernel and small C value is preferred for construction of an SVM to filter spam emails.**

A soft margin SVM with linear kernel and p = 1, C = 0.08 was designed. Verification of the performance was carried out by sampling out 600 feature vectors from the test set. Average accuracy was 92.67%, which is a good result.

Kernel = $K(x_1, x_2) = (x_1^T x_2 + 1)$	Margin = **soft**	p = **1**	C = **0.8**

<u>README:</u> Task_1.m consists of three sections. Kindly run them separately for Task 1. To test the accuracy, run the respective section in Task_2.m after completing Task 1. Please run Task_3.m after adding eval.mat into the folder.